# Ioana Oriana Bercea

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## Positions

1/10/2023–present **Assistant Professor (tenure-track)**,
Division of Theoretical Computer Science
Department of Computer Science, School of Electrical Engineering and Computer Science
KTH Royal Institute of Technology, Stockholm, Sweden

1/05/2021–30/09/2023 **Postdoc**,
Basic Algorithms Research Copenhagen (BARC)
Department of Computer Science
IT University of Copenhagen, Copenhagen, Denmark
Host: Thore Husfeldt

1/09/2018–30/04/2021 **Postdoc**,
School of Electrical Engineering
Tel Aviv University, Tel Aviv, Israel
Host: Guy Even

## Education

03/09/2013–17/08/2018 **Ph.D. in Computer Science**,
University of Maryland, College Park, MD, USA
Thesis: *Approximation Algorithms for Geometric Clustering and Touring Problems*
Advisor: Samir Khuller

30/08/2010–19/05/2013 **M.Sc. in Computer Science**,
University of Maryland, College Park, MD, USA
Advisor: Aravind Srinivasan

25/09/2006–12/06/2010 **B.Sc. in Mathematics (Honors) and B.Sc. in Computer Science**,
University of Chicago, Chicago, IL, USA
Four-year scholarship for tuition, housing and stipend (unconditioned on GPA)

## Honors and Awards

2026 **Best Research Paper Honorable Mention**, SIGMOD

2026 **Research Highlight**, SIGMOD

2025 **Best Research Paper**, VLDB

2023 **Best Artifact**, ACM SIGMOD

2023 **Future Digileader**, Digital Futures

2014–2015 **Outstanding Graduate Assistant Award**, University of Maryland
Top 2% of all UMD Graduate Assistants

2010–2012 **Dean's Fellowship**, University of Maryland

2006–2008, 2009–2010 **Dean's List**, University of Chicago

## Grants

2025 – 2028 **Starting Grant**, Swedish Research Council
Natural and engineering sciences, 1 100 000 SEK/year

## Refereed Conference Articles (Author order alphabetical, unless stated otherwise)

SODA'27 **Hashing for Sampling-Based Estimation – The Missing Lower Tail Bound**
Anders Aamand, Ioana-Oriana Bercea, Jonas Klausen, Jakob Bæk Tejs Houen, Mikkel Thorup
In: *ACM-SIAM Symposium on Discrete Algorithms*, (**SODA**)

NeurIPS'26 **The Heavy Hitter Oracle: Enhancing Frequency Estimation in Skewed Data Streams**
Lisa Schmierer, Ioana-Oriana Bercea *(authors in order of contribution)*
In: *Annual Conference on Neural Information Processing Systems*, (**NeurIPS**)

ESA'26 **Learning-Augmented Online Sorting and TSP**
Ioana-Oriana Bercea, Gerth Stølting Brodal, John Iacono, László Kozma, Debmalya Panigrahi
In: *European Symposium of Algorithms*, (**ESA**)

ICML'26 **Instance-Specific Approximation Ratios for Correlation Clustering and Max-Cut**
Sebastian Lüderssen, Ioana-Oriana Bercea, Stefan Neumannn *(authors in order of contribution)*
In: *International Conference on Machine Learning*, (**ICML**)

SIGMOD'26 **Sublime: Sublinear Error & Space for Unbounded Skewed Streams**
Navid Eslami, Ioana O. Bercea , Rasmus Pagh, Niv Dayan *(authors in order of contribution)*
In: *International Conference on Management of Data*, (**SIGMOD**)
*Best Research Paper Honorable Mention*

VLDB'25 **Diva: Dynamic Range Filter for Var-Length Keys and Queries**
Navid Eslami, Ioana O. Bercea , Niv Dayan *(authors in order of contribution)*
In: *Very Large Data Bases Conference*, (**VLDB**)
*Best Research Paper*
*SIGMOD 2026 Research Highlight*

ICALP'25 **Algorithms for the Diverse-k-SAT problem: the geometry of satisfying assignments**
Per Austrin, Ioana O. Bercea, Mayank Goswami, Nutan Limaye, and Adarsh Srinivasan
In: *EATCS International Colloquium on Automata, Languages, and Programming*, (**ICALP**)
*Invited talk to Highlights of Algorithms (HALG) 2026*

CIAC'25 **Dynamic Filter and Retrieval with One Access to Modifiable Memory**
Ioana O. Bercea, Guy Even, Tomer Even and Gabriel Marques Domingues
In: *International Conference on Algorithms and Complexity*, (**CIAC**)

ESA'24 **Online sorting and online TSP: randomized, stochastic, and high-dimensional**
Mikkel Abrahamsen, Ioana O. Bercea, Lorenzo Beretta, Jonas Klausen, László Kozma
In: *European Symposium on Algorithms*, (**ESA**)

VLDB'24 **Aleph Filter: To Infinity in Constant Time**
Niv Dayan, Ioana O. Bercea , Rasmus Pagh *(authors in order of contribution) (authors in order of contribution)*
In: *Very Large Data Bases Conference*, (**VLDB**)

SWAT'24 **Daisy Bloom Filters**
Ioana O. Bercea , Jakob Bæk Tejs Houen, Rasmus Pagh
In: *19th Scandinavian Symposium and Workshops on Algorithm Theory*, (**SWAT**)

FOCS'23 **Locally Uniform Hashing**
Ioana O. Bercea, Lorenzo Beretta, Jonas Klausen, Jakob Bæk Tejs Houen, Mikkel Thorup
In: *64th IEEE Symposium on Foundations of Computer Science*, (**FOCS**)

SIGMOD'23 **InfiniFilter: Expanding Filters to Infinity and Beyond**
Niv Dayan, Ioana O. Bercea , Pedro Reviriego, Rasmus Pagh *(authors in order of contribution)*
In:*International Conference on Management of Data*, (**SIGMOD**)
*Best Artifact*

STOC'22 **An Extendable Data Structure for Incremental Stable Perfect Hashing**
Ioana O. Bercea, Guy Even
In: *54th Annual ACM Symposium on Theory of Computing*, (**STOC**)

WADS'21 **Dynamic Dictionaries for Multisets and Counting Filters with Constant Time Operations**
Ioana O. Bercea, Guy Even
In: *17th Algorithms and Data Structures Symposium*, (**WADS**)
Invited to the special issue of *Algorithmica*

CIAC'21 **Upper Tail Analysis of Bucket Sort and Random Tries**
Ioana O. Bercea, Guy Even
In: *12th International Conference on Algorithms and Complexity*, (**CIAC**)
Invited to the special issue of *Theoretical Computer Science*

SWAT'20 **A Dynamic Space-Efficient Filter with Constant Time Operations**
Ioana O. Bercea, Guy Even
In: *17th Scandinavian Symposium and Workshops on Algorithm Theory*, (**SWAT**) , pp. 11:1-11:17

APPROX'19 **On the cost of essentially fair clusterings**
Ioana O. Bercea, Martin Groß, Samir Khuller, Aounon Kumar, Clemens Rösner, Daniel R. Schmidt, Melanie Schmidt
In: *22nd International Workshop on Approximation Algorithms for Combinatorial Optimization Problems*, (**APPROX**), pp. 18:1-18:22

CCCG'18 **Improved Bounds for the Traveling Salesman Problem with Neighborhoods on Uniform Disks**
Ioana O. Bercea
In: *30th Canadian Conference on Computational Geometry*, (**CCCG**), pp.129-141

CCCG'16 **Minimizing Uncertainty through Sensor Placement with Angle Constraints**
Ioana O. Bercea, Volkan Isler, Samir Khuller
In: *28th Canadian Conference on Computational Geometry*, (**CCCG**), pp. 287-294

SPAA'14 **On Computing Maximal Independent Sets of Hypergraphs in Parallel**
Ioana O. Bercea, Navin Goyal, David G. Harris, Aravind Srinivasan
In: *26th ACM Symposium on Parallelism in Algorithms and Architectures*, (**SPAA**),pp. 42-50
Invited to the special issue of *TOPC*

## Journal Articles (Author order alphabetical, unless stated otherwise)

2026 **Dynamic Range Filtering Beyond Worst-Case Bounds**
Navid Eslami, Ioana O. Bercea, Niv Dayan *(authors in order of contribution)*
In: *ACM SIGMOD Record*, Volume 55, Issue 1

2022 **Dynamic Dictionaries for Multisets and Counting Filters with Constant Time Operations**
Ioana O. Bercea, Guy Even
In: *Algorithmica*, Special issue on WADS 2021

2021 **Upper Tail Analysis of Bucket Sort and Random Tries**
Ioana O. Bercea, Guy Even
In: *Theoretical Computer Science*, Special issue on CIAC 2021

2016 **On Computing Maximal Independent Sets of Hypergraphs in Parallel**
Ioana O. Bercea, Navin Goyal, David G. Harris, Aravind Srinivasan
In: *ACM Transactions on Parallel Computing*, Special issue on SPAA 2014

## Invited Workshops

Feb. 2026 NII Shonan: *New Era of Data Structures: From Strings to Graphs, and more*
Aug. 2025 Dagstuhl : *Software Performance Engineering*
May 2025 Dagstuhl : *Adaptive and Scalable Data Structures*
May 2025 Dagstuhl : *Learned Predictions for Data Structures and Running Time*
March 2025 NII Shonan: *Algorithmic tools and their applications in emerging models of computation*
July 2024 NU Singapore: *Algorithms and Data Structures Today*

Sept 2023 Bertinoro : *Algorithmic Aspects of Clustering and Related Problems*
May 2023 Dagstuhl : *Scalable Data Structures*
March 2023 NII Shonan : *New Frontiers in Locality in Computation*
Feb. 2021 Dagstuhl : *Scalable Data Structures*
Oct. 2019 IGAFIT: *Workshop for Postdoctoral Researchers in Algorithms*
Sept. 2019 GI-Dagstuhl : *Algorithms for Big Data*
March 2019 Dagstuhl : *Theoretical Foundations of Storage Systems*
Jan. 2019 Dagstuhl: *Data Structures for the Cloud and External Memory Data*
Oct. 2018 *7th French-Israeli Workshop on Foundations of Computer Science*

## Invited Talks

June 2026 Randomness in Computing, University of Sheffield, UK
Dec 2025 A Celebration of TCS, Sydney, Australia
Oct 2025 WAVE (Women in Algorithms, Venture into Exploration), Copenhagen, Denmark
Aug. 2025 TU Wien
June 2025 Bocconi Theory Day, Milan, Italy
March 2025 Tokyo Tech
Dec 2024 Australasian Summer School on Recent Trends in Algorithms, Sydney, Australia
July 2023 Romanian Algorithms Days, virtual

## PhD supervision

2026 – **Simone Moretti**
KTH Royal Institute of Technology

2026 – **Fabien Niessen**
KTH Royal Institute of Technology
primary supervisor: Aris Gionis

2025 – **Riccardo Zuech**
University of Padova
co-supervised with Francesco Silvestri

2021-2025 **Jonas Østergaard Klausen**
University of Copenhagen
co-supervised with Mikkel Thorup, Jacob Holm
→ postdoc at MPI Saarbrücken

## Teaching

Fall 2023 – **Course Responsible, Teacher for DD2440 Advanced Algorithms, MsC**,
KTH Royal Institute of Technology, Department of Computer Science

Summer 2022 & 2023 **Co-Intructor for Algorithms and Data Structures BSc, MSc**,
IT University of Copenhagen, Department of Computer Science

Spring 2016 **Co-Intructor for CMSC 122: Intro to Computer Programming via the Web**,
University of Maryland, Department of Computer Science

## Service

2028 **Dagstuhl seminar co-organizer** *Simplicity and Efficiency of Data Structures (28221)*
2026– **Steering Committee** *IGAFIT Highlights of Algorithms*
2026 **Conference organizer** *Highlights of Algorithms (HALG 2026), A celebration of complexity*
2025 **Program Committee co-chair** SOSA 2025 *(with Rasmus Pagh)*
**Program Committee member**

2026 ESA (track S), EuroPar, SEA, COCOON, SWAT, ICALP, STACS, SODA

2025 ESA (track A)

2024 RANDOM, ISAAC, SOSA

2020 ICDCN

2024 – 25 **Guest journal editor** Theory of Computing *(Special issue on RANDOM 2024)*

2024 – **Co-editor** Algorithms Column for the Bulletin of EATCS *(with Thomas Erlebach)*

2015–2016 **Graduate Student Representative**,
University of Maryland, Computer Science Department Council

2012–2014 **Graduate Student Representative**,
University of Maryland, Computer Science Department Education Committee